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Guidelines for the Aerodynamic Design of sCO_2 Centrifugal Compressor Stages

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This paper explores the viability of established criteria to design centrifugal compressor stages that operate CO_2 close to the critical point. The work is split into two complementary studies. At first, several stage configurations are generated by varying characteristic design parameters, such as the inlet and outlet flow coefficient, work coefficient, and stage Mach number. Then, three selected configurations are optimized with a surrogate-assisted evolutionary strategy to refine specific geometrical parameters, which control the meridional shape and splitter location. The stage aerodynamics and performance are assessed with a validated RANS solver based on the homogeneous equilibrium assumption for the two-phase mixture. One of the key findings of the parametric studies is that designing a stage with a flow coefficient smaller than that suggested by design practices prevents the occurrence of two-phase flows. The onset of a secondary phase at the impeller intake alters the fluid compressibility, producing density gradients across saturated condition that can cause flow separation. Such separation occurs for liquid-like CO_2 but not for vapor-like CO_2, implying that consolidated design rules can apply to the latter case notwithstanding the presence of condensing flows. Shape optimizations reveal that modest efficiency improvements can be gained by changing the meridional channel and splitter location. Moreover, the optimization problem is highly multimodal, as more than 30 different geometries can yield equivalent design efficiency. As a side effect of the optimization, the rangeability is significantly modified. Specifically, increasing the inlet slope angle of the shroud contour extends the operating range at high flow rates but reduces it at low flow rates. Therefore, this geometrical parameter can be controlled to adjust the stage rangeability without compromising design efficiency. [DOI: 10.1115/1.4063311]

1 Introduction

Closed power cycles that use supercritical carbon dioxide (sCO_2) as a working fluid have the potential to be more efficient than organic- and steam-based cycles for various temperature ranges and applications [1]. However, their market deployment has been hindered by the unique challenges posed by the system components, which have been requiring significant investment to develop and operate proof-of-concept prototypes [2].

One of the most critical components is the main centrifugal compressor, which has to compress CO_2 close to the thermodynamic critical point. The near-critical compression poses specific challenges that do not emerge in conventional compressors: (i) significant gradients in CO_2 thermodynamic properties, preventing an ideal-gas description [3], and (ii) possible two-phase flows due to local expansions below saturation [4]. Recent experimental activities on MW-scale sCO_2 compressors have demonstrated the importance of these effects on compressor performance maps [5,6]. Therefore, revising existing design criteria has become a priority to promote the technology growth and prevent future setbacks.

The first step in aerodynamic design is selecting a suitable compressor architecture that meets design constraints (structural, cost, space, etc.) and achieves the desired performance targets (required pressure ratio, minimum operating range, efficiency, etc.). This selection step is carried out based on a few key parameters that can summarize stage performance. To retrieve these design parameters, most designers rely on seemingly universal statistical diagrams, e.g., the Balje diagram. Early design examples of sCO_2 compressors made no exceptions [7,8]. For centrifugal compressors, alternative selection variables have historically replaced Balje's specific diameter and specific speed, such as the flow coefficient ϕ_{01} and the work coefficient λ_d. Based on proprietary performance data, different authors have agreed that centrifugal compressors should be designed with $\phi_{01} = 0.08 - 0.10$ to achieve maximum efficiency [9,10].

In essence, all selection criteria are built on historical performance data of conventional compressors and leverage the similitude results. The straight application of such design guidelines to sCO_2 compressors appears questionable and may lead to suboptimal designs. Therefore, the first part of this work aims to establish the link between a few characteristic design variables and sCO_2 stage performance in terms of efficiency, rangeability, and susceptibility to two-phase flows. The analysis, in the form of multiple parametric studies, exploits a validated computational fluid-dynamic (CFD) solver that makes use of generalized thermodynamics and can allow homogeneous two-phase flows at thermodynamic equilibrium. The three-dimensional flow path is obtained by specifying similar distributions for the meridional channel, blade angle, and thickness

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Manuscript received August 1, 2023; final manuscript received August 3, 2023; published online September 29, 2023. Editor: Jerzy T. Sawicki.

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among different compressor designs. In designing a sCO_2 stage, Pelton et al. [11] stated that they performed multiple CFD iterations to refine the flow path in order to maximize the efficiency and range, but they did not provide details or intermediate outcomes. Other authors did not clearly report these pieces of information either [12,13]. Assigning these distributions properly can be a challenging task, as there are no clear guidelines available. Therefore, in the second part of this work, an evolutionary surrogate-assisted shape optimization method is applied to different stages in order to systematically optimize the meridional shape and the splitter location. The overall goal is to discover possible optimal trends and provide guidelines for the aerodynamic design of sCO_2 compressor stages.

2 Geometry Generation

In this section, the workflow to obtain the three-dimensional geometry of a centrifugal compressor stage starting from a few characteristic design variables is detailed. The archetypal stage consists of a shrouded impeller, which is structurally feasible thanks to the low peripheral velocity ($< 200\,\mathrm{m/s}$), and a vaneless diffuser, which grants a larger operating range [11]. Other elements, such as the volute, crossover bend, and return channel, are not included in the analyses because they will depend on overall compressor arrangements and specific applications (number of stages required, stage arrangement on the shaft, possible integrally geared layout, etc.). The design parameters are the flow coefficient ϕ_{01}, the outlet flow coefficient ϕ_2, the work coefficient λ_d, and the stage Mach number M_u, together with the intake thermodynamic state $(P_{01}, T_{01} \mapsto h_{01}, s_1)$ and the mass flowrate $\dot{m}$. Fluid properties are calculated by interrogating the thermophysical property database COOLPROP [14].

At first, a mean-line approach sets the main geometrical quantities (radii, blade heights, blade angles). The combination of specified ϕ_{01} and M_u provides the outlet impeller radius R_2 and the rotational speed ω. The inlet section is designed as follows:

(1) The selection of the hub radius R_{1h} is subordinated to the shaft design. The shaft has to withstand the induced torsion by the torque and is sized accordingly

$$R_{\mathrm{sh}} = \sqrt[3]{\xi_1\xi_2\xi_3 \frac{2\sqrt[3]{3}}{\pi} \frac{(\dot{m}_d \lambda_d \omega R_2^2)}{\sigma_a}} \tag{1}$$

where $\xi_1(=2)$ is a safety factor that accounts for bending moments and fatigue, $\xi_2(=1.5)$ accounts for the increased torque during startup, and $\xi_3(=1.3)$ accounts for a maximum torque larger than the design target. The allowable stress σ_a is set at $685\,\mathrm{N/mm^2}$, which is the minimum yield stress for 39NiCrMo3 steel with $20\,\mathrm{mm} < R_{\mathrm{sh}} < 50\,\mathrm{mm}$. Then, R_{1h} is increased from R_{sh} by 10% for the impeller assembly.

(2) The shroud radius R_{1s} is obtained by minimizing the corresponding relative Mach number $M_{w_{1s}}$. The minimization task is carried out numerically with the differential evolution algorithm [15]. Each $M_{w_{1s}}$ evaluation requires (i) solving the continuity equation for the meridional velocity v_{m1} with a first-guess static density ρ_1, (ii) computing the static state $h_1 = h_{01} - 0.5\,v_1^2$ with the assumption of no swirl ($v_1 = v_{m1}$), (iii) iterating on (i) and (ii) while updating $\rho_1 = \rho(h_1, s_1)$, and (iv) computing the relative velocity $w_{1s} = \sqrt{v_1^2 + (\omega R_{1s})^2}$.

(3) The hub diameter is increased to comply with the specified sweep angle γ_s while keeping the same flow-passage area.

(4) As the three-dimensional impeller shape is generated via the ruled-surface approach, only the hub and shroud blade angles have to be specified. The shroud blade angle is set equal to the flow angle that resulted from the Mach number minimization ($\hat{\beta}_{1s} = \beta_{1s}$). The flow angle is also computed at the root-mean-square radial location. Then, these two flow angles are linearly extrapolated along the radial coordinate to determine the hub blade angle $\hat{\beta}_{1h}$. Consequentially, the incidence is null in the upper part of the impeller, where most of the diffusion occurs, while the hub blade profile tolerates a small negative incidence ($\hat{\beta}_{1h} - \beta_{1h} < -10\,\mathrm{deg}$).

As per the outlet section, λ_d and ϕ_2 entirely shape the velocity triangle. The outlet blade angle $\hat{\beta}_2$, which is equal to the hub and shroud blade profile, is determined by including the flow deviation via the Wiesner correlation [16, Eq. 5.36]. The outlet blade height b_2 is computed from the continuity equation, assuming an isentropic efficiency for the impeller in the calculation of the outlet static density. This isentropic efficiency can be estimated with semi-empirical loss correlations. Nonetheless, an error of five percentage points in the efficiency changes b_2 by only $\sim 1\%$ thanks to the small CO_2 compressibility in the thermodynamic conditions of interest. Hence, the meanline efficiency is set at the constant value of 0.95 to determine b_2. The axial length L_{ax} is estimated with the Aungier's correlation [9, Eq. 6-3].

Once the main geometrical parameters are settled, the meridional channel, the blade angle distribution, and the thickness distribution can be defined accordingly. Four Bezier control points generate the shroud and the hub meridional contours. The two extreme control points pass through the inlet and outlet coordinates already determined, while specified angles δ_{ij}, $i = \{1,2\}, j = \{h,s\}$, set the two internal control points as in Fig. 1. The blade angles are distributed along the meridional coordinate as suggested by Aungier [9]

$$\hat{\beta}_s = \hat{\beta}_{1s} + (\hat{\beta}_2 - \hat{\beta}_{1s})(3\tilde{m}^2 - 2\tilde{m}^3) \tag{2}$$

$$\hat{\beta}_h = \hat{\beta}_{1h} + A\tilde{m} + B\tilde{m}^2 + C\tilde{m}^3 \tag{3}$$

$$A = -4(\hat{\beta}_2 - 2\bar{\beta}_h + \hat{\beta}_{1h}) \tag{3a}$$

$$B = 11\hat{\beta}_2 - 16\bar{\beta}_h + 5\hat{\beta}_{1h} \tag{3b}$$

$$C = -6\hat{\beta}_2 + 8\bar{\beta}_h - 2\hat{\beta}_{1h} \tag{3c}$$

$$\bar{\beta}_h = (1-K)(\hat{\beta}_2 + \hat{\beta}_{1h})/2 \tag{3d}$$

The shroud and hub profiles are stacked along the leading edge following the specified lean angle γ_l. The parameter K in Eq. (3*d*) is changed to meet the prescribed rake angle γ_r at the trailing edge.

The thickness is added orthogonally to the camber line. The blade is thinner at the leading edge, while it has a constant thickness after a fraction of the meridional coordinate (see Fig. 1). The leading edge has a circular shape. The trailing edge is cutoff at R_2.

The splitter blade derives from the main blade by cutting the main-blade camber line at a specified meridional coordinate $\tilde{m}_{\mathrm{spl}}$. Then, the thickness distribution, as reported in Fig. 1, is applied to the cut camber line.

The vaneless diffuser extends up to $R_4 = 1.6R_2$. A front pinch is designed to bring the absolute flow angle to 65 deg at $R_3 = 1.2R_2$ to avoid flow instabilities in the vaneless diffuser at part-load conditions. The pinch is reproduced by a third-order curve that preserves the continuity of the shroud and of its first derivative at R_2 and becomes straight at R_3.

3 Parametric Investigation

The effect of parameters ϕ_{01}, ϕ_2, λ_d, and M_u on compressor design is examined. For these parametric studies, the inlet total conditions are $P_{01} = 80\,\mathrm{bar}$ and $T_{01} = 305.15\,\mathrm{K}$, i.e., supercritical CO_2 lies in the liquid-like domain ($s_1/s_c < 1$). Liquid-like conditions result from the optimization of sCO_2 closed power systems as a consequence of the reduction in compression work [17]. However, other applications, such as carbon capture and sequestration, can demand the compressor to operate in the vapor-like region [18].

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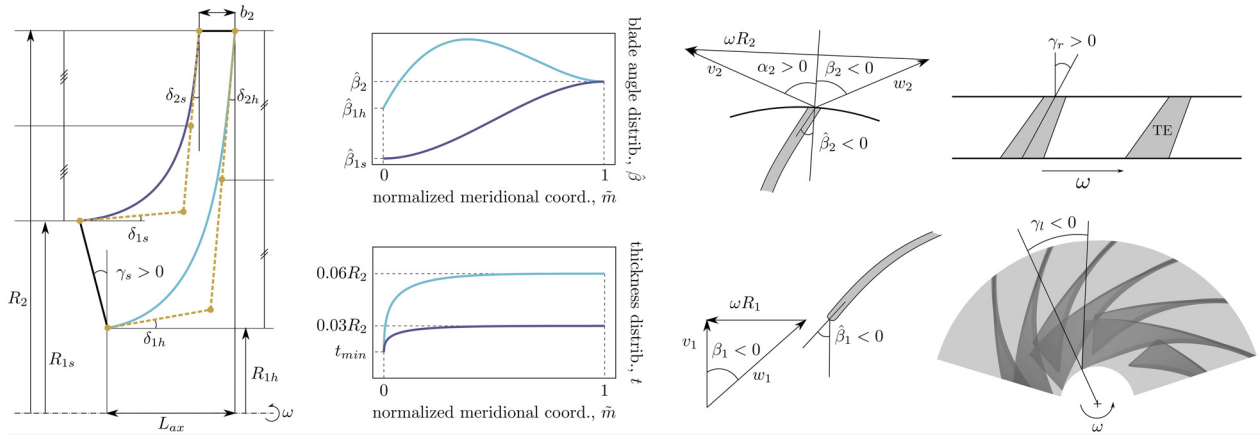


Fig. 1 Impeller meridional channel (left), blade angle, and thickness distribution (middle left), velocity triangles and blade angles at impeller trailing edge and leading edge (middle right), rake, and lean angle definition (right)

The effect of vapor-like conditions on stage design will be analyzed in Sec. 3.4 separately.

The mass flowrate determines the compressor size and, therefore, the importance of nonscalable geometrical features (roughness and fillets) but does not alter the design. As the focus of this work is on compressor design, a constant value $\dot{m}_d = 100.0\,\mathrm{kg/s}$ is selected unless otherwise specified. For the same reason, the effect of roughness and fillets is also neglected.

The overall blade number is 16 from loading considerations [9, Eq. 6-22], of which eight main blades and eight splitter blades starting from $\tilde{m}_{\mathrm{spl}} = 0.20$. Sweep, lean, and rake angle, as well as slope angles for the meridional channel, are fixed among different designs. Some of these values will be rediscussed in the light of optimization outcomes in Sec. 4. Table 1 sums up all constant values assumed for the following parametric analyses.

The compressor aerodynamics and performance are assessed by means of steady Reynolds-averaged Navier–Stokes simulations. The pressure-based flow solver was purposely developed for sCO_2 compressors by Romei et al. [19] as it accounts for two-phase flows and accurate equations of state. Thermophysical properties are invoked using pressure and specific enthalpy as state variables, assuming thermodynamic equilibrium between phases. The two phases also move with the same velocity. The equation of state [20] and correlations for transport properties [21,22] are incorporated in the flow solver by means of look-up tables, whose resolution is discussed in the original Ref. [19]. The turbulence is introduced by the two-equation shear-stress transport model [23]. Advective terms are discretized using the QUICK scheme with a flux limiter [24]. Diffusion terms are discretized with the second-order central-difference scheme. The gradients are computed with the least square method. The pressure is interpolated at cell faces with second-order accuracy. The computational method was validated on a representative sCO_2 flashing flow in Ref. [19]. Subsequently, it was also verified against the numerical predictions of the barotropic model [25], which was validated against experimental data of a MW-scale sCO_2 compressor stage [6], finding equivalent numerical predictions in terms of total pressure, efficiency, and choked flowrate.

Table 1 Constants for parametric studies

Constant	Value	Unit
$\dot{m}_d$	100.0	kg/s
t_{min}	0.6	mm
γ_s, γ_l, γ_r	15, 0, 0	deg
δ_{1s}, δ_{2s}	0, 5	deg
δ_{1h}, δ_{2h}	10, 0	deg
m_{spl}	0.20	
N_{bl}, N_{spl}	8, 8	

A single-blade passage including both the main and the splitter blade is modeled by exploiting periodic boundary conditions. Sealings and corresponding leakage flows on the disk and cover disk are not simulated but modeled separately as postprocessing. The computational domain extends from one axial length upstream of the leading edge (inlet domain) to the end of the vaneless diffuser ($R_4/R_2 = 1.6$). All walls are smooth and adiabatic. The hub and shroud walls of the inlet domain and of vaneless diffuser are stationary, while the hub and shroud of the impeller are rotating to model a shrouded stage. The main blade and splitter blade are rotating as well. The inlet boundary conditions are total pressure and total enthalpy. Moreover, turbulent intensity and the eddy-viscosity ratio are set at 5% and 10, respectively, to comply with the two-equation turbulence model. Finally, the mass-flowrate is specified as the outlet boundary condition unless otherwise specified.

The meshes are generated with ANSYS-TURBOGRID and are composed of 3.3×10^6 hexahedral elements. A new mesh is produced every time the geometry changes. The first-cell y^+ is between one and five for both blades and endwalls. Fifteen cells are placed in the boundary layer (up to $y^+ \sim 500$). Such geometrical discretization allows for avoiding wall functions in turbulence modeling. The blade span is discretized with 28 elements, yielding 57,000 elements in the blade-to-blade plane. An example of the resulting mesh is displayed in Fig. 2.

This mesh setup is the result of a dedicated grid-independent study on a reference compressor geometry ($\phi_{01} = 0.05$, $\phi_2 = 0.20$, $\lambda_d = 0.65$, and $M_u = 0.60$). In addition to the previous mesh, a coarser and a finer mesh with 1.3 and 8.9×10^6 elements, respectively, are also generated. The three meshes share the same boundary layer discretization and have a refinement ratio of 1.4 (rounded down) for the elements along the blade span. Two key

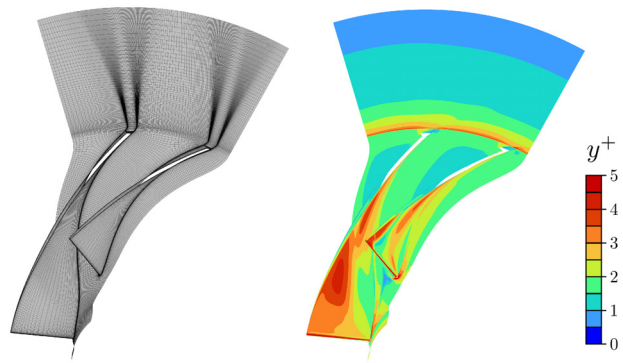


Fig. 2 Example of mesh adopted for the following studies and associated y^+ distribution

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quantities are selected for the grid study, namely, the work coefficient $\lambda = l/(\omega R_2)^2$ and the aerodynamic efficiency $\eta_{aero} = l_p/l$, where l is obtained from the overall torque on rotating elements and l_p is computed from mass-weighted averages of total pressure and total enthalpy at diffuser outlet. With reference to the medium mesh, the grid convergence indexes [26] are 1.2% for λ and 0.1% for η_{aero}.

3.1 Influence of ϕ_{01} and ϕ_2. The effect of ϕ_{01} and ϕ_2 on compressor design and potential cross-interactions between these two parameters are investigated in this section. As illustrated in Fig. 3, a small ϕ_{01} leads to a stage that features a large impeller outlet radius and, consequentially, a smaller rotational speed to achieve the target outlet peripheral velocity. On the contrary, a high ϕ_{01} leads to a more compact design with a larger axial-to-radial curvature and rotational speed. ϕ_2 has a twofold effect: at first, it establishes the magnitude of the meridional velocity and, thus, the blade height, and then it affects the flow angle leaving the impeller. Therefore, a large ϕ_2 implies a smaller blade height and a more meridional flow direction at the impeller outlet, i.e., $\alpha_2 = \arctan(\lambda_d/\phi_2)$, hence reducing the need for a diffuser pinch.

In this parametric study, ϕ_{01} varies from 0.02 to 0.08 with a step of 0.01. The maximum value of ϕ_{01} is selected to avoid excessive rotational speeds for MW-scale shafts ($< 30,000$ rpm). For each value of ϕ_{01}, three ϕ_2 are considered, i.e., 0.20, 0.25, and 0.30. As a result, 21 different designs are generated by means of the algorithm detailed in Sec. 2. To isolate the effect of ϕ_{01} and ϕ_2, the other design parameters are kept constants, namely, $\lambda_d = 0.65$ and $M_u = 0.60$.

Figure 4 illustrates the effect of ϕ_{01} and ϕ_2 on the aerodynamic efficiency η_{aero}. The maximum efficiency has a plateau for ϕ_{01} between 0.04 and 0.07, where η_{aero} achieves a value between 0.960 and 0.962. For this range of ϕ_{01}, ϕ_2 affects the efficiency slightly with a variation of 0.1 percentage points between 0.20 and 0.25. Designs with $\phi_2 = 0.30$ are always suboptimal. The efficiency reduces when $\phi_{01} < 0.04$ and $\phi_{01} > 0.07$, albeit the decay is modest (0.2 percentage points). The efficiency penalty becomes significant at $\phi_{01} = 0.02$, for which $\eta_{\text{aero}} = 0.945$. Such high values of η_{aero} should not surprise the reader as η_{aero} only accounts for the aerodynamic losses that occur in the flowpath (flow separations and smooth-wall skin friction), which are generally small in sCO_2 compressors by virtue of the large Reynolds number ($> 10^8$ for analyzed geometries, using the camber line as reference length).

To clarify how ϕ_{01} affects aerodynamic losses, Fig. 5 reports the mass-weighted averaged loss coefficient on the meridional plane for all ϕ_{01} under study at their optimal ϕ_2. For low flow coefficients, the large wetted surface generates high skin-friction losses. Moreover, as the flow channel is narrow, the boundary layers occupy a significant fraction of the channel height ($\sim 55\%$ for $\phi_{01} = 0.02$ at R_4). The importance of this viscous loss decreases as ϕ_{01} increases.

Another equally important contribution to overall loss is the separation that occurs at the shroud toward the impeller outlet. The entity of this separation becomes significant at high-flow

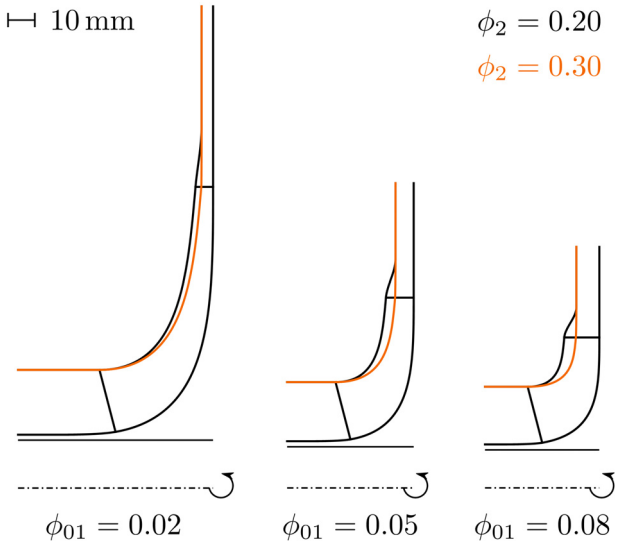


Fig. 3 Effect of ϕ_{01} and ϕ_2 on meridional shape

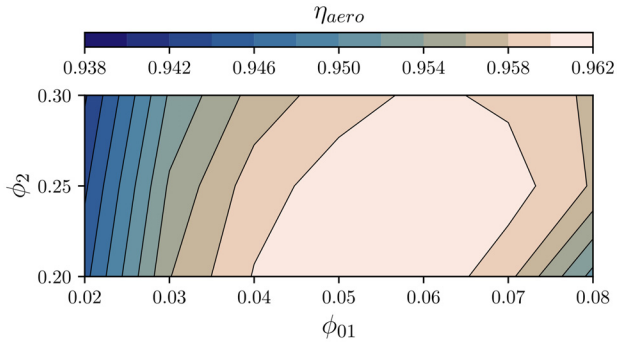


Fig. 4 Effect of ϕ_{01} and ϕ_2 on aerodynamic efficiency ($\lambda_d = 0.65$, $M_u = 0.60$, $P_{01} = 80$ bar, $T_{01} = 305.15$ K)

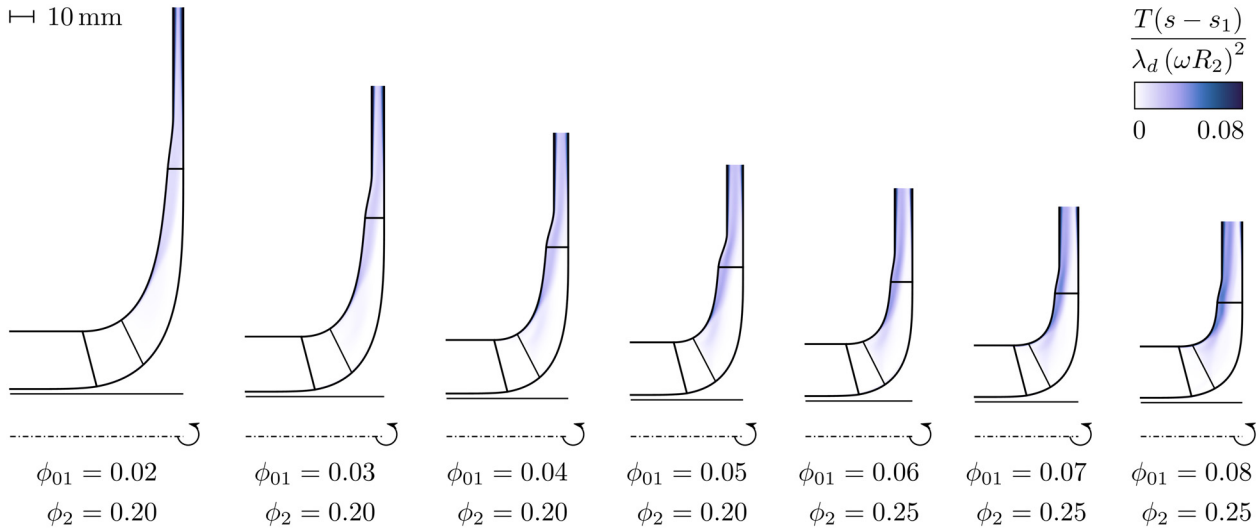


Fig. 5 Loss distribution on the meridional plane for stages with different ϕ_{01}

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coefficients for two reasons: first, the higher axial-to-radial curvature promotes the separation, and, second, the flow is decelerated more. Notwithstanding that, all impellers exchange the same amount of work with the fluid ($-0.021 < 1 - \lambda/\lambda_d < 0.005$), the way in which the work is exchanged is substantially different. At low flow coefficients, the flow diffusion is small ($w_2/w_{1s} = 0.91$ for $\phi_{01} = 0.02$) and the centrifugal action is mainly responsible for the work exchanged. Conversely, the flow diffusion becomes significant at high flow coefficients ($w_2/w_{1s} = 0.59$ for $\phi_{01} = 0.08$) to compensate for the smaller centrifugal action. Moreover, as long as ϕ_{01} increases, the importance of two-phase effects increases as well, promoting flow separation as discussed later in the paper.

From the previous discussion, it can be concluded that different ϕ_{01} can be selected without a great sacrifice in aerodynamic efficiency. However, the overall stage efficiency has also to account for various sources of parasitic work, such as the extra work to recompress recirculating shroud leakage flow (leakage loss) and to overcome the frictional force in clearance gaps (windage loss). These additional works are not explicitly simulated but are modeled with proper correlations. Accordingly, stage efficiency is defined as

$$\eta_{\text{stage}} = \frac{\dot{m}\, l_p}{(\dot{m} + \dot{m}_l) l + 0.5(C_{Md} + C_{Mcd})\rho_2 \omega^3 R_2^5} = \frac{\eta_{\text{aero}}}{1 + \dfrac{\dot{m}_l}{\dot{m}} + \dfrac{1}{8}\dfrac{\rho_2}{\rho_{01}}\dfrac{(C_{Md} + C_{Mcd})}{\phi_{01}\lambda}} \tag{4}$$

where $\dot{m}_l/\dot{m} \approx 0.0005/\phi_{01}$ [16, Eq. 10.51], and C_{Md} and C_{Mcd} are computed as proposed by Aungier [9, Eq. 4-31], assuming smooth walls and a gap-to-outlet radius of 0.04.

Figure 6 illustrates the impact of parasitic works and their weight relative to aerodynamic losses. The contribution of parasitic losses increases at low flow coefficients, where they become comparable with the aerodynamic losses. The optimal ϕ_{01} for stage efficiency is 0.07, although a variation of ± 0.01 results in an efficiency drop of only 0.2 percentage points. At low flow coefficients, the efficiency drop becomes appreciable and exceeds 1.0 percentage points when $\phi_{01} < 0.04$. Between windage and leakage loss, the former is found to be predominant for all flow coefficients.

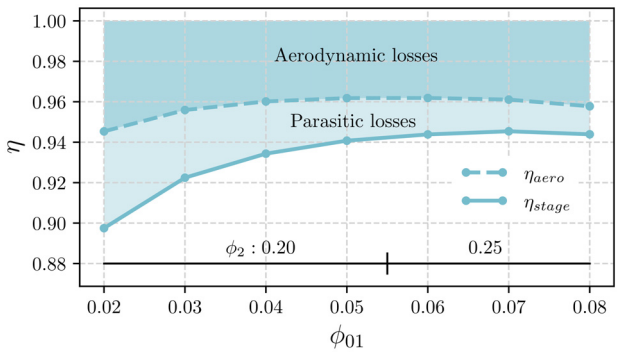


Fig. 6 Change of stage efficiency with ϕ_{01}

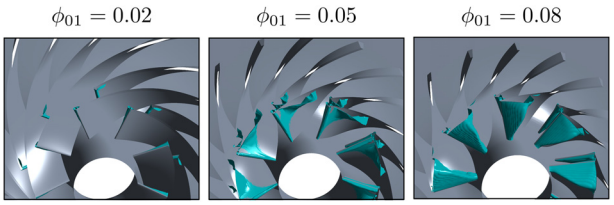


Fig. 7 Change of two-phase region with ϕ_{01}

It is worth highlighting that η_{stage} is representative of the single stage alone. To derive the overall compressor efficiency, the number of stages and their arrangement on the shaft should be decided first, and then more sources of loss have to be included accordingly, such as those occurring in the intake duct, crossover bends, return channels, and volutes.

Alongside efficiency, there are other factors that can favor the choice of a specific design. As the compression starts close to the saturation curve, local-two phase flows can originate at the impeller intake [17]. Besides the uncertainty of compressing a two-phase flow mixture, the authors have experimentally demonstrated that two-phase flows are responsible for a significant reduction in the operating range [6]. Hence, a design that minimizes the risk of two-phase flows can be desirable. To this end, Fig. 7 highlights the two-phase region for three archetypal stages with different ϕ_{01} at corresponding optimal ϕ_2. The displayed two-phase region, which is identified by the isosurface of vapor mass fraction equal to 0.001, grows with ϕ_{01}. For $\phi_{01} = 0.02$, the amount of two-phase flow is limited and concentrated near the leading edge. As previously stated, at low flow coefficients the aerodynamic loading is small because of the high centrifugal action, and the most significant flow accelerations arise from the flow turning around the leading edge. By increasing the flow coefficient, the aerodynamic loading increases as well, and the corresponding acceleration along the suction side becomes sufficient to let a larger portion of the flow achieve saturation conditions.

Another important feature of a compressor stage is the operating range, which is delimited by the *surge limit* at low flow rates and by the *choking limit* at high flow rates. The surge limit is determined by progressively reducing the mass flowrate at the outlet section until the total-to-static pressure ratio P_4/P_{01} starts decreasing [16]. Then, the maximum P_4/P_{01} is identified using a bisection method with a tolerance in the mass flowrate of 1%. To identify the choking limit, the outlet static pressure is specified and progressively decreased until the change in mass flowrate is below 1%.

The evolution of the operating range with ϕ_{01} is displayed in Fig. 8. The trend is opposed to the one of the efficiency, featuring a larger rangeability at low flow rates. Quantitatively, the operating range, expressed as a difference between choked and surge mass flowrate normalized to the design one, is 61% for $\phi_{01} = 0.02$ but reduces to 28% for $\phi_{01} = 0.08$. The main responsible for this difference is a significant variation in the surge limit (from $0.56\,\dot{m}_d$ to $0.86\,\dot{m}_d$), which can be conveniently correlated with the amount of flow diffusion. Hence, a centrifugal stage designed with a high value of w_2/w_{1s}, which means low ϕ_{01}, can be preferred for applications where operability at part-load conditions is requested.

The trend of the choking limit, which varies between $1.12\,\dot{m}_d$ and $1.17\,\dot{m}_d$, is less clear because it results from two competing effects: for low ϕ_{01}, the fraction of two-phase flows, which sets the sonic throat because of the small speed of sound [6], is limited, but the throat area is reduced as well. As a result, the choking limit is slightly affected by ϕ_{01}.

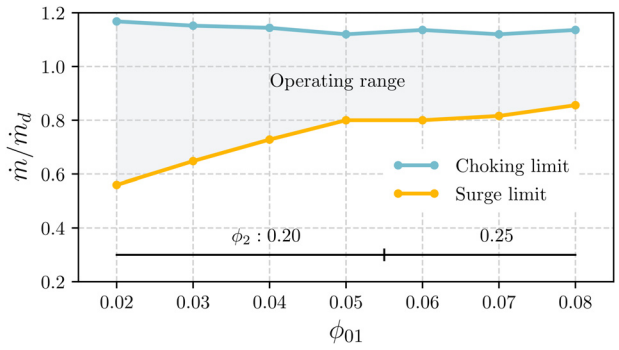


Fig. 8 Change of operating range with ϕ_{01}

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3.2 Influence of λ_d. The study in Sec. 3.1 is repeated by changing λ_d, which passes from 0.65 to 0.60. A smaller λ_d yields a more meridional flow at the impeller outlet in the same spirit of an increase in ϕ_2. $\lambda_d > 0.65$ would require a vaned diffuser to recover static pressure efficiently and, hence, it is excluded from this analysis. M_u is set at 0.6, and the same range and discretization of Sec. 3.1 are considered for ϕ_{01} and ϕ_2.

The first consequence of reducing λ_d is that the fluid receives less work from the impeller. However, this work reduction has a minor impact on the pressure rise P_{04}/P_{01}, which passes from 2.3 when $\lambda_d = 0.65$ to 2.2 when $\lambda_d = 0.60$. As discussed in Sec. 3.3, the major contribution to the pressure rise for the present operating conditions is provided by a variation in M_u rather than λ_d.

The results in terms of η_{aero} are collected in Fig. 9(*a*). By comparison with Fig. 4, the trend of η_{aero} with ϕ_{01} and ϕ_2 is the same regardless of the change in λ_d. The optimal ϕ_2 does not vary, and it is 0.20 for $\phi_{01} < 0.05$ and 0.25 otherwise. Overall, the efficiency is reduced by up to 0.3 percentage points when λ_d reduces to 0.60 from 0.65. The same considerations also apply for η_{stage}, with a slight increase in windage losses because of the λ_d reduction—see Eq. (4). Quantitatively, the windage-loss increase yields a reduction in η_{stage} of additional 0.1 percentage points.

By inspecting the flow fields, the two-phase region is barely affected by the change in λ_d, and the same considerations expressed in Sec. 3.1 also applies in this circumstance. The operating ranges of stages having $\phi_{01} = 0.02$ and 0.08 are simulated and compared with those obtained with $\lambda_d = 0.65$. The nearly invariant two-phase region leads to a similar choking limit, which changes by less than 1% in both cases. The margin from surge slightly improves because of the smaller diffusion associated with the lower λ_d. Quantitatively, the minimum mass flow rates reduce by 3.7% and 5.6% for $\phi_{01} = 0.02$ and 0.08, respectively, when $\lambda_d = 0.65 \mapsto 0.60$.

As the blade loading decreases with a lower value of λ_d, it is worthwhile to consider a stage configuration with fewer blades. Therefore, the previous study is repeated by designing impellers with seven main blades and seven splitter blades. The comparison of the two contour maps presented in Fig. 9 demonstrates that the impeller with a smaller number of blades is either less efficient or equally efficient as the baseline configuration with 8 + 8 blades. The reduction in η_{aero} is limited to 0.3 percentage points at low ϕ_{01} and does not alter the qualitative trend of η_{aero} with respect to ϕ_{01} and ϕ_2.

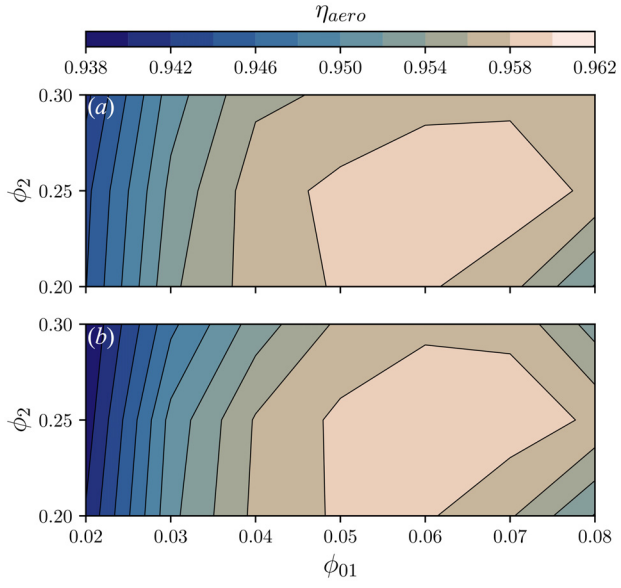


Fig. 9 Effect of ϕ_{01} and ϕ_2 on aerodynamic efficiency ($\lambda_d = 0.60$, $M_u = 0.60$, $P_{01} = 80$ bar, $T_{01} = 305.15$ K). Note that λ_d is changed with respect to Fig. 4: (*a*) number of blades: 8 + 8 (identical to Fig. 4) and (*b*) number of blades: 7 + 7.

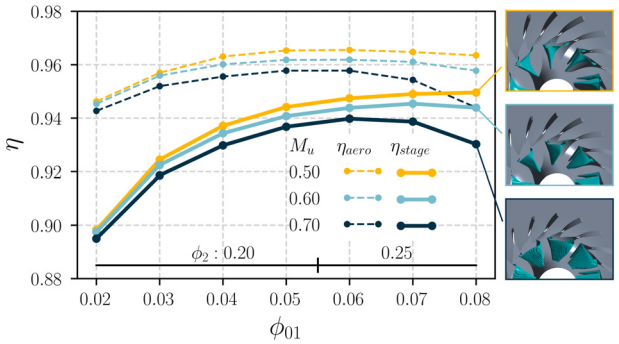


Fig. 10 Effect of M_u on stage efficiency

3.3 Influence of M_u. The effect of M_u is isolated by comparison with results in Sec. 3.1 for different ϕ_{01} at corresponding optimal ϕ_2. λ_d is set at 0.65 accordingly. A change in M_u implies a new stage geometry as $R_2 = \sqrt{\dot{m}/(4\rho_{01}c_{01}M_u\phi_{01})}$ from ϕ_{01} definition. The analysis is carried out for two M_u, namely, 0.50 and 0.70, in addition to previous designs obtained with $M_u = 0.60$. As already illustrated in Ref. [17], an increase in M_u promotes the onset of two-phase flows, which is responsible for choked-flow conditions. As further evidence of this, few geometries choke for $\dot{m} < \dot{m}_d$ when $M_u = 0.75$, thus restricting the following analyses to $M_u \leq 0.70$.

The impact of M_u on efficiency grows with ϕ_{01}, as illustrated in Fig. 10. As a matter of fact, Mach numbers are less than 0.6 for the single-phase flows, and small variations around this value should not affect the stage performance (negligible compressibility effects). Nonetheless, the significant drop in speed of sound at saturation leads to a supersonic flow regime for the two-phase mixture. The increase in M_u slightly alters the maximum Mach number but raises the flow fraction that becomes supersonic because of the two-phase transition. This explanation is consistent with the previous evidence for which stage geometries designed with high ϕ_{01} are more susceptible to two-phase flows. As a consequence, the aerodynamic efficiency is barely affected by M_u at low ϕ_{01}, while the drop becomes significant at high ϕ_{01}.

This efficiency penalty is not linear with M_u: for example, for $\phi_{01} = 0.08$ the drop in efficiency is 0.6 percentage points when $M_u = 0.50 \mapsto 0.60$ and becomes 1.4 percentage points when $M_u = 0.60 \mapsto 0.70$. For this latter case, the design flowrate is close to the choked-flow condition, and the two-phase flow occupies almost the entire blade channel. The efficiency drop from η_{aero} to η_{stage} is the same for all M_u, suggesting that parasitic losses are practically insensitive to the M_u choice. By examining Eq. (4), the Mach number enters only in ρ_2/ρ_{01}, but such dependency is weak due to the small CO_2 compressibility in these thermodynamic conditions.

Finally, M_u has a significant impact on the pressure rise when CO_2 behaves like a liquid. For this specific case, P_{04}/P_{01} is 1.8 when $M_u = 0.5$, 2.3 when $M_u = 0.6$, and 2.7 when $M_u = 0.7$. Therefore, although low-M_u stages feature higher efficiency and a reduced amount of two-phase flows, more stages would be required to deliver the overall pressure ratio required by the application.

3.4 Influence of Fluid Properties. As a final parametric study, the effect of fluid properties on stage design is investigated. To highlight the associated implications on compressor design, in this section compressor stages are designed for the exemplary intake thermodynamic state $P_{01} = 74$ bar and $T_{01} = 305.15$ K ($s > s_c$) and are compared with stage geometries of Sec. 3.1 obtained for $P_{01} = 80$ bar and $T_{01} = 305.15$ K ($s < s_c$). λ_d and M_u are fixed between the two studies and are 0.65 and 0.60, respectively. Because of different fluid compressibility, identical λ_d and M_u yield a smaller pressure rise ($P_{04}/P_{01} = 1.4$) across the stage for the vapor-like condition. ϕ_{01} and ϕ_2 are combined as in Sec. 3.1 to produce 21 additional stages tailored for the vapor-like thermodynamic condition. $\dot{m}_d$ is

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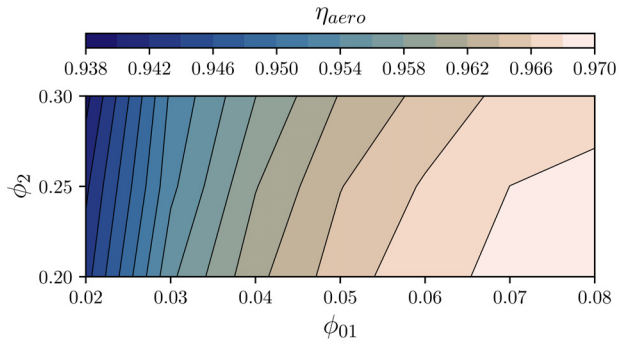


Fig. 11 Effect of ϕ_{01} and ϕ_2 on aerodynamic efficiency ($\lambda_d = 0.65$, $M_u = 0.60$, $P_{01} = 74$ bar, $T_{01} = 305.15$ K). Note that P_{01} is changed with respect to Fig. 4 as well as the maximum value of η_{aero}.

changed to 48.3 kg/s to match the volumetric flowrate $\dot{m}_d/\rho_{01}$ of Sec. 3.1 and ensure similar size effects.

The aerodynamic efficiency for the combination of ϕ_{01} and ϕ_2 assuming vapor-like condition is reported in Fig. 11. The comparison between this latter figure and Fig. 4 reveals two main differences. At first, the maximum efficiency when CO_2 behaves like a vapor is 0.8 percentage points higher than the liquid-like counterpart. Then, the optimal ϕ_{01} region shifts toward high ϕ_{01} values, namely, $0.07 - 0.08$ (to be compared to $0.04 - 0.07$ for the liquid-like case). Moving to the η_{stage}, the parasitic works modeled in Eq. (4) slightly increase in magnitude due to the rise of 3.5% in ρ_2/ρ_{01} for the vapor-like condition. The overall impact of parasitic works does not change between the two thermodynamic conditions and is mainly driven by ϕ_{01}. It follows that $\phi_{01} = 0.08$ maximizes η_{stage}, which achieves the value of 0.955. $\phi_{01} = 0.08$ is also an optimal value for standard centrifugal stages [10], suggesting that consolidated design principles presumably apply for compressor stages that operate vapor-like CO_2. Then, η_{stage} decreases by decreasing ϕ_{01}: the efficiency drop is 0.6 percentage points for $\phi_{01} = 0.06$ and approximately doubles for any additional 0.01 reduction in ϕ_{01}.

The inspection of the flow field reveals why compressor stages for the liquid-like condition are less efficient and require a smaller ϕ_{01} than their vapor-like counterparts. To this end, Fig. 12 reports the limiting streamlines on the splitter blade for the stages designed with $\phi_{01} = 0.08$ and $\phi_2 = 0.25$. The smaller efficiency in the liquid-like condition is caused by flow separation just downstream of the two-to-single phase transition point. On the contrary, the two-phase region in the vapor-like condition does not alter streamlines and hence does not produce any flow separation. Such different fluid-dynamic behavior is triggered by the different volumetric evolution

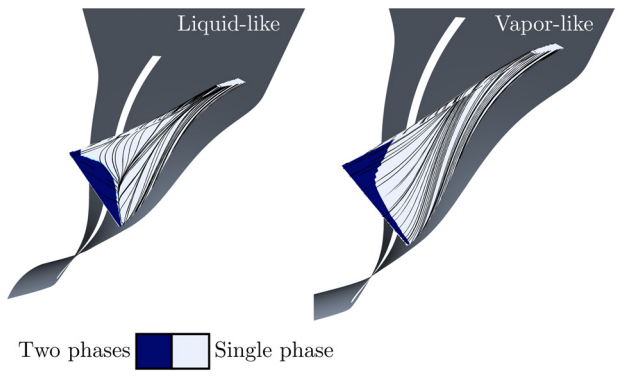


Fig. 12 Limiting streamlines on compressor splitter blade ($\phi_{01} = 0.08$, $\phi_2 = 0.25$, $\lambda_d = 0.65$, $M_u = 0.60$) designed for liquid-like ($P_{01} = 80$ bar, $T_{01} = 305.15$ K) and vapor-like ($P_{01} = 74$ bar, $T_{01} = 305.15$ K) conditions

of the two multiphase mixtures. On the one hand, in the liquid-like case, the dispersed phase (vapor bubbles) has a large specific volume and then considerably reduces the bulk density. Hence, the flow accelerates more, requiring a recompression via a shock-wave that makes the flow separate. On the other hand, in the vapor-like case, the dispersed phase (liquid droplets) tends to occupy less volume, yielding a mixture density that changes as the single-phase vapor-like CO_2. Therefore, no shocks are required to reestablish the imposed flow compression.

In conclusion, stages designed for liquid-like conditions benefit from lower ϕ_{01} than conventional stages because of the reduction in ϕ_{01} reduces the extent of the two-phase region and consequent flow separation.

4 Shape Optimization

The parametric studies conducted so far have involved stage dimensionless parameters. A few geometrical parameters were assumed in generating the three-dimensional shape, see Table 1, based on literature recommendations [9,16] and authors' personal experience with sCO_2 compressors. In this section, dedicated shape optimizations are performed to refine these initial choices and understand whether there is a margin for improvement and how to select these geometrical parameters properly.

With reference to Table 1, γ_l and γ_r are not included in the shape optimization because they are strictly related to the stage structural integrity and have little effect on aerodynamic performance [27]. δ_{2h} is set at 0 to avoid unfeasible converging-diverging meridional channels when $\delta_{2h} > \delta_{2s}$. The remaining geometrical parameters are included in the optimization and are $\Gamma = \{\gamma_s, \delta_{1h}, \delta_{1s}, \delta_{2s}, \tilde{m}_{spl}\}$. The optimization problem aims at maximizing the aerodynamic efficiency while constraining the work coefficient and reads as

$$\max_{\Gamma} \quad \eta_{aero} \tag{5a}$$

$$\text{subject to} \quad |\lambda - \lambda_d| < 0.01 \tag{5b}$$

The constrained optimization problem is solved with the genetic algorithm (GA). As the GA requires many evaluations to converge, a surrogate strategy is adopted to reduce the computational cost, leveraging validated libraries from the open-source Dakota toolkit [28].

Figure 13 outlines the optimization workflow. At first, 25 stage geometries are generated within the prescribed design space with Latin hypercube sampling. The corresponding η_{aero} and λ are calculated with CFD simulations as detailed in Sec. 3. Two distinct Kriging (also known as Gaussian process) are introduced as surrogates of the CFD simulation, and their means conditional to the simulated geometries are used to reproduce the functional dependencies $\Gamma \mapsto \eta_{aero}$ and $\Gamma \mapsto \lambda$. The employed Kriging models have up to second-order monomials of the input variables as the

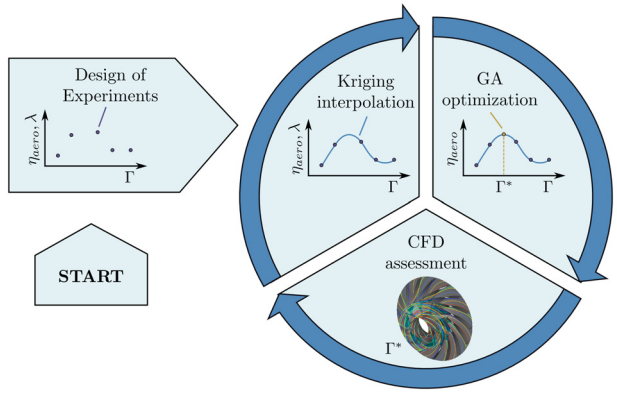


Fig. 13 Flow chart summarizing the surrogate-assisted shape optimization strategy

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Table 2 Design space for optimization studies

Variable	Lower value	Upper value	Unit
γ_s	0	30	deg
δ_{1h}	0	20	deg
δ_{1s}	0	20	deg
δ_{2s}	0	15	deg
m_{spl}	0.10	0.40	

mean trend and incorporate a Gaussian correlation function. The Kriging hyperparameters are determined by maximum likelihood estimation. The readers interested in the mathematical formulation and properties are referred to the specialized literature (see, e.g., Ref. [29]).

Once the two Kriging predictors are obtained, the GA optimization is executed on surrogate responses. The so-identified optimal stage geometry is assessed with CFD, and new Kriging predictors are regenerated on all simulated cases, namely, initial design of experiments plus predicted optimal geometries by surrogate optimization. In this way, the surrogate reliability progressively increases by adding geometries close to the expected optimal region. The optimization procedure is repeated until convergence, i.e., the surrogate optimization converges to the same optimal geometry of the previous iteration. Alternatively, the optimization will stop after 75 iterations, setting a maximum of 100 CFD simulations for each optimization.

Three optimizations are performed, targeting stages with $\phi_{01} = 0.02, 0.05$, and 0.08. The corresponding ϕ_2 are selected according to optimal values of Sec. 3.1. λ_d and M_u are set at 0.65 and 0.60, respectively. The design space, equal for the three optimizations, is reported in Table 2.

4.1 Optimized Stage Designs. Stage geometries that maximize η_{aero} while satisfying the constraint (5b) are summarized in Table 3. The increase in efficiency is modest and amounts to 0.3 percentage points for $\phi_{01} = 0.02$ and 0.2 percentage points for both $\phi_{01} = 0.05$ and 0.08. This result arguably suggests either the meridional channel and splitter location have a minor impact on design performance or the baseline geometries are near optimal.

To verify this latter statement, the optimized design variables are compared with corresponding baseline values. The optimal γ_s is close to 20 deg for the three optimized stages, and it is slightly higher than the initial value (15 deg). Also, δ_{2s} is near the baseline value of 5 deg, indicating that optimal stages are characterized by radial meridional velocities at the trailing edge. The splitter blades are moved downstream compared to the baseline configurations, but the displacement is reduced with ϕ_{01}. As explained in Sec. 3.1, the diffusion is small for low-ϕ_{01} stages, hence the optimization removes a part of the splitter blade to reduce the wetted surface and the associated skin-friction loss.

Table 3 Baseline and optimized geometries and performance

Variable	Base	Opt	Base	Opt	Base	Opt	Unit
Design							
ϕ_{01}	0.02		0.05		0.08		
ϕ_2	0.20		0.20		0.25		
M_u			0.60				
λ_d			0.650				
Optimized							
γ_s	15	22	15	22	15	18	deg
δ_{1h}	10	0	10	15	10	11	deg
δ_{1s}	0	18	0	20	0	0	deg
δ_{2s}	5	4	5	4	5	2	deg
m_{spl}	0.20	0.27	0.20	0.27	0.20	0.23	
Output							
λ	0.664	0.657	0.650	0.646	0.647	0.641	
η_{aero}	0.945	0.948	0.962	0.964	0.958	0.960	
η_{stage}	0.898	0.900	0.941	0.943	0.944	0.946	

On the contrary, trends for optimal δ_{1h} and δ_{1s} are not clearly recognizable. For $\phi_{01} = 0.02$, the curvature on shroud contour reduces ($\delta_{1s} = 18$ deg), while the opposite occurs for the hub curvature ($\delta_{1h} = 0$ deg). The trend is reversed for $\phi_{01} = 0.08$, while both curvatures reduce for $\phi_{01} = 0.05$. To shed light on such design aspect, all stage geometries simulated during the optimization that exhibit η_{aero} greater than or equal to the baseline value and satisfy λ_d-constraint are plotted in Fig. 14. It turns out that many equivalent geometries (given the modest efficiency improvement) can be obtained by modifying the meridional channel and splitter location. Out of 100 simulated geometries for each optimization task, 43, 64, and 32 geometries that perform equally or slightly better than the baseline are obtained for $\phi_{01} = 0.02, 0.05$, and 0.08, respectively.

Collecting all these designs, some geometrical limits to achieve optimal efficiency can be identified by taking the most restricting extreme out of the three optimizations: (i) 10 deg $\leq \gamma_s \leq$ 25 deg, (ii) 6 deg $\leq \delta_{1h} \leq$ 20 deg, (iii) 0 deg $\leq \delta_{1s} \leq$ 20 deg, (iv) $\delta_{2s} <$ 5 deg, and (v) $0.17 \leq \tilde{m}_{\text{spl}} \leq 0.32$.

Although many geometries can yield comparable design performance, the stage rangeability may differ. To have a better insight into this latter open point, the efficiency curves over the flow range for baseline and optimized stages are reported in Fig. 15. For $\phi_{01} = 0.08$, the flow range does not change, and efficiency curves basically overlap with minor improvements for the optimized configuration (limited to 0.2 percentage points). Instead, significant variations in the flow range are observed for $\phi_{01} = 0.02$ and 0.05. The optimized stages exhibit a range increase at high flow rates by 16% and 10% and a range reduction at low flow rates by 8% and 4% for $\phi_{01} = 0.02$ and 0.05, respectively. Both variations can be explained in light of the increase in δ_{1s} for the optimized stages, which implies a larger throat area and a smaller shroud meridional

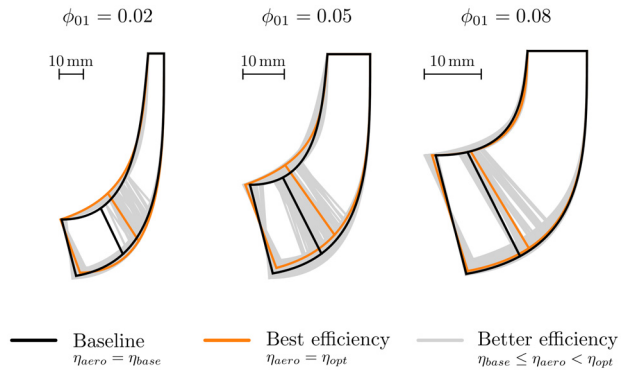


Fig. 14 Stage geometries simulated during shape-optimization processes that perform better than corresponding baselines

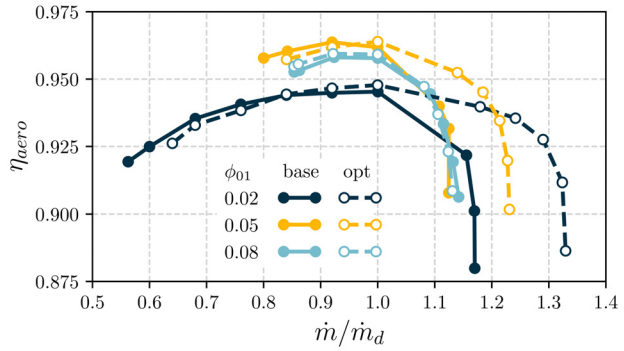


Fig. 15 Efficiency curves over the flow range for baseline and optimized stage geometries

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length, as one can infer from Fig. 14. On the one hand, the throat areas are increased by the same relative percentages as the ones observed in the choking limits, thus simply improving the swallowing capacities of the optimized stages. On the other hand, the reduction in the shroud meridional lengths raises the blade loading, thus facilitating the occurrence of reversed flows as the flowrate reduces.

5 Conclusion

This work has examined the possible choices for the aerodynamic design of centrifugal compressor stages that operate with supercritical carbon dioxide (sCO_2). Distinct parametric analyses have shed light on the individual role of stage characteristic parameters, namely, inlet and outlet flow coefficient (ϕ_{01}, ϕ_2), work coefficient (λ_d), and stage Mach number (M_u), on efficiency, rangeability, and susceptibility to two-phase flows. Shape optimizations applied to different stage configurations have allowed us to determine optimal meridional channel evolution and splitter location. Notably:

- ϕ_{01} primarily affects stage geometry and performance. Low ϕ_{01} (as low as 0.02) implies lower efficiency but larger operating range, the opposite for high ϕ_{01} (up to 0.08). The fraction of flow undergoing phase transition increases with ϕ_{01}.
- ϕ_2 and λ_d have small influence on efficiency (~ 0.3 percentage points), hence they can be safely selected based on the desired velocity triangle at the trailing edge and possible requirements for the downstream diffuser.
- M_u has a large impact on the onset of two-phase flows. The suggestion is to rely on multiple stages to keep $M_u \leq 0.60$. Far from the saturation curve, the occurrence of two-phase flows is less probable, and M_u can be selected based on established design practices.
- Many meridional channel shapes can provide equivalent design performance. Common threads are the almost straight radial part at the impeller outlet and splitter blades starting at around 25% of the meridional coordinate. The curvature of the shroud contour can be adjusted to shift the operating range toward low or high flow rates.

Based on these findings, a number of recommendations can be drawn. If the main target concerns design performance, a high-ϕ_{01} stage compatible with possible constraints on rotational speed is recommended. This choice ensures maximum efficiency but a limited operating range. Moreover, the stage has to handle a significant fraction of the flow undergoing two-phase transition, with all associated consequences (e.g., possible early deterioration, operational uncertainty). Otherwise, if the target is stable and reliable operation, a low-ϕ_{01} stage is recommended. In the authors' opinion, this is the best option for sCO_2 compressors for closed power cycles because of the small sensitivity of net cycle efficiency on compressor efficiency. For this purpose, $\phi_{01} = 0.04$ may represent a suitable compromise between efficiency and rangeability. For vapor-like conditions, the importance of two-phase flows decreases, hence conventional design methods can be utilized.

Finally, the curvature of the shroud contour can be adjusted depending on the off-design requirements. A smaller curvature is indicated to widen the operating range at large flow rates at the expense of that at low flow rates.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

b = blade height
c = speed of sound
C_{Md} = torque coefficient for disk
C_{Mcd} = torque coefficient for cover disk
CFD = computational fluid dynamic
GA = genetic algorithm
h = mass-specific enthalpy
l = specific work
l_p = specific polytropic work
L_{ax} = impeller axial length
m = meridional coordinate
$\dot{m}$ = mass flow rate
$\tilde{m}$ = normalized meridional coordinate
$\dot{m}_l$ = shroud-leakage mass flow rate
$\tilde{m}_{spl}$ = initial meridional coordinate for splitter
M_u = stage Mach number, $(\omega R_2)/c_0$
M_w = relative Mach number, w / c
N_{bl} = main blade number
N_{spl} = splitter number
P = pressure
R = radius
R_{sh} = shaft radius, Eq. (1)
s = mass-specific entropy
sCO_2 = supercritical carbon dioxide
t = thickness
T = temperature
t_{min} = minimum thickness
v = absolute velocity
v_m = meridional velocity
w = relative velocity
y^+ = dimensionless wall distance
α = absolute flow angle
β = relative flow angle
$\hat{\beta}$ = blade angle
γ_l = lean angle
γ_r = rake angle
γ_s = sweep angle
Γ = vector of geometrical variables for optimization
δ = meridional channel slope angle
η_{aero} = aerodynamic (polytropic) efficiency, l_p/l
η_{stage} = stage efficiency, Eq. (4)
λ = calculated work coefficient, $l/(\omega R_2)^2$
λ_d = imposed work coefficient during design
ξ_i = safety factor, $i = 1, 2, 3$
ρ = density
σ_a = allowable stress
ϕ_{01} = flow coefficient, $\dot{m}_d/(4\rho_{01}\omega R_2^3)$
ϕ_2 = outlet flow coefficient, $v_{m2}/(\omega R_2)$
ω = rotational speed
$(\cdot)_{base}$ = baseline geometry
$(\cdot)_c$ = at thermodynamic critical point
$(\cdot)_d$ = at design condition
$(\cdot)_h$ = hub
$(\cdot)_m$ = meridional
$(\cdot)_{opt}$ = optimized geometry
$(\cdot)_s$ = shroud
$(\cdot)_{0i}$ = total state at i section
$(\cdot)_1$ = impeller inlet
$(\cdot)_2$ = impeller outlet
$(\cdot)_3$ = end of vaneless diffuser pinch
$(\cdot)_4$ = vaneless diffuser outlet

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